Tanya Singh

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EDUCATION

University of Maryland, College Park **College Park, MD** **Graduation Date: May 2027**

Bachelor of Science in Computer Science, Minor in General Business *GPA: 3.985, Dean's List*

- Relevant Coursework: Object-Oriented Programming, Applied Probability and Statistics, Discrete Structures, Computer Systems, Linear Algebra, Organization of Programming Languages, Algorithms, Data Science, Advanced Data Structures, Machine Learning, Compilers
- Leadership: Committee Member at Association for Women in Computing, Marketing Lead at Google Developer Student Club, Committee Head at Campus Coders Crew, Researcher in the First-Year Innovation & Research Experience

SKILLS

- **Languages:** Python, Java, C, SQL, R, OCaml, Rust, JavaScript, HTML/CSS, MATLAB, x86 Assembly, Haskell, Racket, Dafny
- **Libraries:** Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn
- **Tools & Platforms:** Microsoft 365 Copilot Agent Builder, Git/GitHub, Azure DevOps, ServiceNow, Microsoft Excel, VS Code, RStudio
- **Concepts:** Machine Learning (Classification, Clustering, Regression), Large Language Models, Prompt Engineering, AI Agent Development, Knowledge Base Design, Hypothesis Testing, Data Visualization, Data Analysis, Agile/Scrum, Zero Trust
- **Certifications:** Microsoft Office Specialist (Associate)

WORK EXPERIENCE

Navy Federal Credit Union **Vienna, VA** **May 2026 - August 2026**

Summer Associate

- Deployed an M365 Copilot-powered Zero Trust AI Assistant that provides cited answers on 29 active projects, 45 capabilities, and 152 activities.
- Designed a metadata-driven knowledge base by cataloging 84 Zero Trust documents and standardizing AI retrieval across teams and work-streams.
- Reduced AI response times by 90%+ by cutting retrieval latency from 2+ minutes to 10 seconds through prompt engineering and optimization.

Lavner Education **Washington, DC** **June 2025 - August 2025**

Intern/Instructor

- Instructed 120+ students in coding, STEM & Medicine, All-Girls STEAM, robotics, and more through interactive, age-appropriate lessons.
- Mastered a hands-on curriculum to deliver a one-on-one learning experience to 40+ students across a range of STEM subjects.
- Led set-up and maintenance of on-site check-in and computer systems, increasing efficiency by 20%.

Easy Dynamics **McLean, VA** **January 2025 - January 2025**

Intern

- Evaluated the ability of 4+ LLMs to help researchers find, collect, and summarize information on payroll threats and fraud during research tasks.
- Designed 12+ research prompts to compare model outputs, analyzing each LLM's strengths and weaknesses on accuracy and completeness.
- Developed and presented a proposal on findings and recommendations for using LLMs in this research, built with PowerPoint, Excel, and Word.

Code Ninjas **Leesburg, VA** **November 2022 - November 2023**

Code Sensei

- Taught 50+ students aged 5 - 14 to write and understand code in programming languages such as Scratch, JavaScript, C#, and more.
- Applied the center's curriculum to guide students through 5+ software platforms such as Roblox Studio, MCreator, and Unity.

PROJECTS

Machine Learning Stroke Risk Prediction | *Python* **2026**

- Developed an end-to-end stroke prediction pipeline in Python using Pandas, NumPy, and Scikit-learn on 5,110 patient records. Applied preprocessing, feature engineering, and scaling to compare Random Forest and Logistic Regression models, achieving 81% stroke-class recall and analyzing age, glucose, and BMI as predictive features.

Lexer, Parser, and Interpreter | *OCaml* **2025**

- Engineered an OCaml-like dynamically typed programming language by building a full lexer, recursive-descent parser, and interpreter. Applied language design principles, grammar-based parsing, lexical scoping, closure formation, and environment-based evaluation, reinforcing core concepts in programming language implementation.

Performance Optimization & Multithreading | *C* **2025**

- Designed and implemented a custom dynamic memory allocator in C to explore explicit memory management and system-level optimization. Applied multithreading with pthreads to build a matrix normalizer that benchmarked allocator performance, addressing concurrency, synchronization, and cache efficiency in computational workloads.

Personal Portfolio | *HTML/CSS, JavaScript* **2025**

- Created a fully responsive and interactive personal website from scratch using HTML/CSS and JavaScript, deployed on a custom domain. The website features smooth animations, a filterable projects page, and separate pages for experience, research, coursework, and contact to showcase professional experiences.

RESEARCH

Effectiveness of Large Language Models (LLMs) in Weather Forecasting for Arlington, VA | *R* **2025**

- Co-authored a research paper evaluating Large Language Models (GPT-5, Claude Sonnet 4, and Gemini 2.5 Pro) at forecasting 92 days of Arlington, VA weather from one year of historical data. Applied prompt engineering, cleaned and compiled model outputs, and ran paired t-tests and Wilcoxon signed-rank tests in R with effect sizes ($\alpha = 0.05$); all three models differed significantly on high temperatures ($p < 0.001$), while GPT-5's low-temperature predictions showed no statistically significant difference from observed temperatures ($p = 0.66$).